VGGRPO: Towards World-Consistent Video Generation with 4D Latent Reward

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Research Questions

- How can video diffusion models be aligned for geometric consistency without architectural modifications or expensive RGB-space reward computation?
- Can geometry foundation models be effectively connected to video diffusion latents to enable 4D scene-aware post-training?
- How can latent-space reinforcement learning improve both camera motion smoothness and cross-view geometric coherence in generated videos?

Core Idea

Video diffusion models produce visually impressive videos but suffer from geometric inconsistencies like camera jitter and structural drift. Existing solutions either modify the architecture (compromising generalization) or use RGB-based rewards (expensive due to VAE decoding). VGGRPO's key insight is to "stitch" a geometry foundation model directly to the diffusion model's latent space via a lightweight connector, enabling geometry prediction without RGB decoding. This allows efficient Group Relative Policy Optimization with two complementary rewards: camera smoothness (penalizing jittery motion) and reprojection consistency (enforcing 3D coherence across views). The approach works because it leverages pretrained geometric priors while operating in the computationally efficient latent space.

Key Formula

Latent Geometry Model

$$\hat{\Phi}_\psi = \Phi_{\hat{\ell}+1:L} \circ S_\psi$$

\hat{\Phi}_\psi is the latent geometry model, \Phi_{\hat{\ell}+1:L} are the downstream layers of the geometry foundation model from layer \hat{\ell}+1 to L, S_\psi is the learned 3D convolutional connector with parameters \psi that maps VAE latents to intermediate geometry model features. This enables geometry prediction directly from latents without RGB input.

Camera Motion Smoothness Reward

$$r_{motion}(z_0) = \frac{1}{2}\left(\frac{1}{1+e_{trans}(z_0)} + \frac{1}{1+e_{rot}(z_0)}\right)$$

r_{motion} is the camera motion smoothness reward, z_0 are the denoised latents, e_{trans} and e_{rot} are translational and rotational smoothness errors based on scale-normalized accelerations. The reward approaches 1 for smooth trajectories and decreases toward 0 as jitter increases, encouraging stable camera motion.

Key Figure

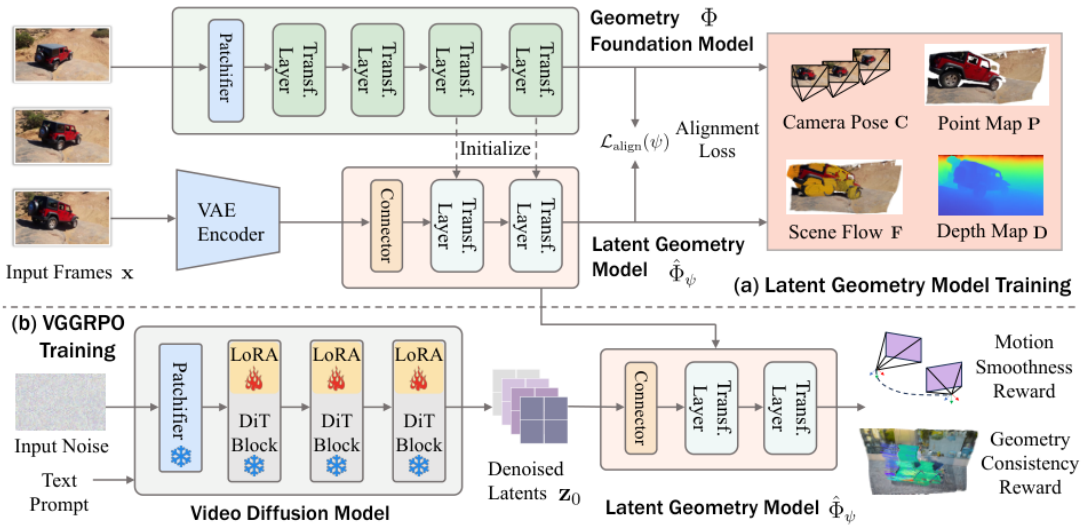

Figure 2 | **Method Overview.** (a) **Latent Geometry Model.** We connect latents from the diffusion VAE encoder to a geometry foundation model via a lightweight connector, yielding a Latent Geometry Model that predicts 4D scene geometry directly from video latents. (b) **VGGRPO training.** We perform latent-space GRPO using two complementary rewards, camera motion smoothness and geometry reprojection consistency, computed entirely in latent space with the latent geometry model. Together, these components align the video diffusion model toward 4D world-consistent generation on both static and dynamic scenes.

Figure 2: Shows the complete VGGRPO framework architecture with both the Latent Geometry Model training (connecting VAE latents to geometry foundation models) and the VGGRPO training process using complementary rewards in latent space.

Limitations

- Requires additional training of the Latent Geometry Model connector, adding complexity to the overall pipeline
- Performance depends on the quality of the underlying geometry foundation model used for stitching
- May show reduced Dynamic Degree scores due to explicit camera smoothing that reduces optical flow magnitude
- Limited to geometry foundation models that can be effectively connected via the stitching approach